

5. Then status Work Order has been **CLOSE**



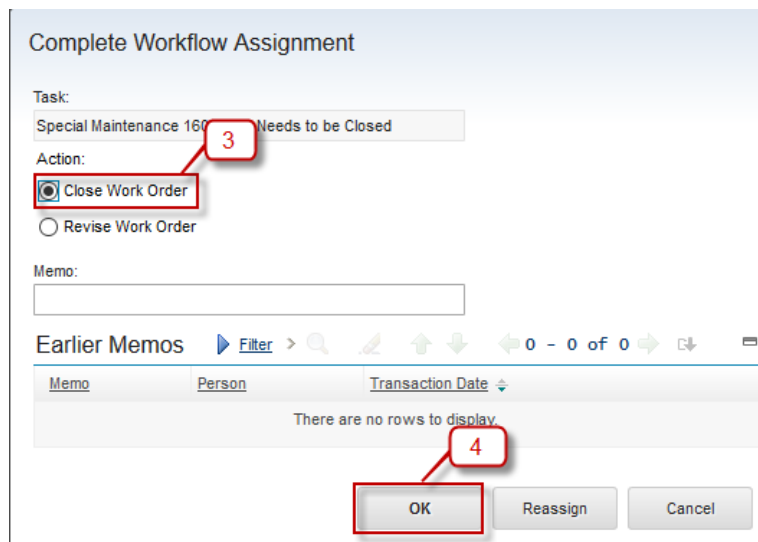
Inbox / Assignments

Description

Next Assignment Due: 5/10/16 9:41 PM

Special Maintenance 16000691 Needs to be Closed

1 - 1 of 1



Complete Workflow Assignment

Task:
Special Maintenance 16000691 Needs to be Closed

Action:
☒ Close Work Order
☐ Revise Work Order

Memo:

Earlier Memos

Filter > 0 - 0 of 0

Memo	Person	Transaction Date
There are no rows to display		

OK Reassign Cancel

1.3 Exercises

2 Preventive Maintenance (PM) Work

2.1 PM Work Overview

Preventive maintenance (PM) work is a maintenance work generated by PM schedule in **Preventive Maintenance** application, to maintain assets or locations in satisfactory operating condition by providing for systematic inspection, detection, and correction of incipient failures either before they occur or before they develop into major defects. The works include tests, measurements, adjustment, parts replacement, and cleaning.

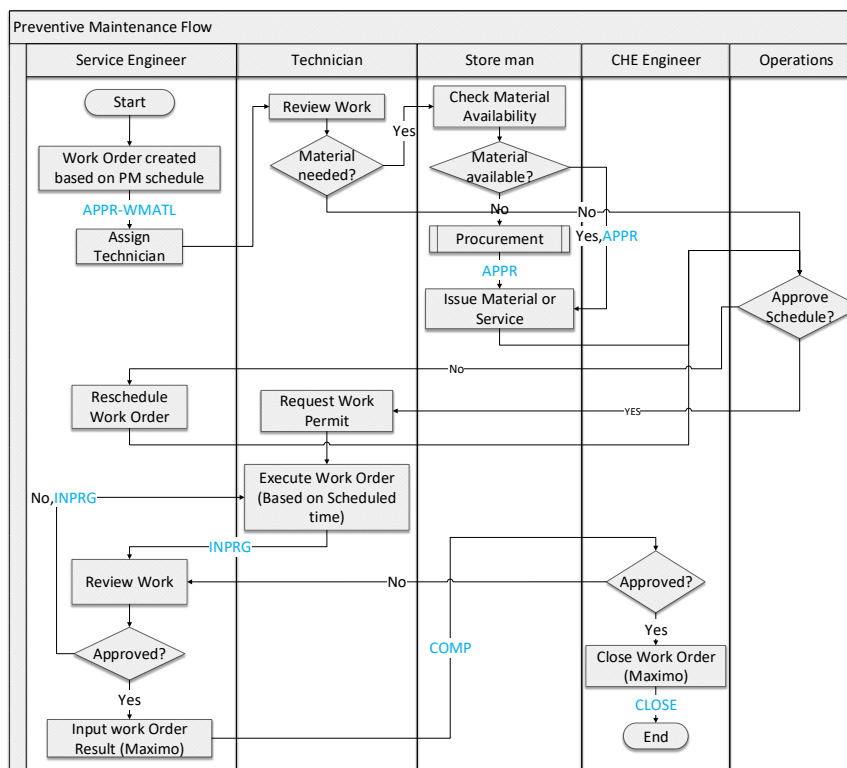
PM work order tends to follow planned guidelines from time-to-time to prevent equipment and machineries breakdown. The guidelines (task to be performed, resource needed, etc.) is contained in **Job Plan** that registered in PM schedule. When PM schedule generate a PM work order, the registered job plan will be applied automatically into the work order.

PM work requires downtime of the assets to be performed. Therefore, the schedule need to be considered more carefully, to minimize the impact on business process. When the schedule of PM

work order generated by PM schedule does not fit, a schedule revision can be done towards those PM work order.

2.2 PM Work Cycle

Below image show how Preventive Maintenance work will be managed in EMS.



Below are detail scenario for Preventive Maintenance (PM) work order:

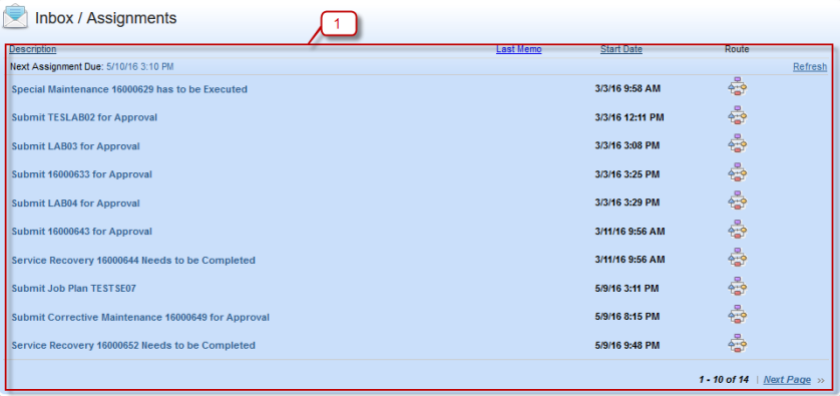
1. PM work order created automatically based on PM schedule defined in **Preventive Maintenance** application, with the initial status is **APPR**.
2. Planned material of the work order can be issued from the storeroom. If the materials are unavailable, procurement should be held to purchase them.
3. An approval assignment will be sent to *Operation*, whether the work is allowed to be executed in the scheduled time or not. *Service Engineer* must revise the schedule if *Operation* reject the scheduled time.
4. *Service Engineer* reviews the work result and input the result into EMS. A closing approval assignment will be sent to *CHE Engineer*, whether the work order can be closed or not.

2.2.1 PM Work Order Submission

As stated above, PM Work Order generated based on PM schedule defined in **Preventive Maintenance** application. When a PM Work Order is generated, an assignment will appears in *Service Engineer's* Start Center. *Service Engineer* needs to review the work order, especially its schedule, before submitting.

Below are step by step for PM Work Order submission:

1. Click Description of the Assignment to go to related Work Order



2. Input **Schedule Start** and **Schedule Finish**
3. Check the **"Requires Asset Downtime?"** checkbox if it's not checked yet



4. Click (Route Workflow)

Scheduling Information

Target Start:

Actual Start:

Target Finish:

Actual Finish:

Scheduled Start:

Duration:

Scheduled Finish:

Predecessors:

Start No Earlier Than:

Include Tasks in Schedule?

☒

Finish No Later Than:

Requires Asset Downtime?

☐

Follow-up Work

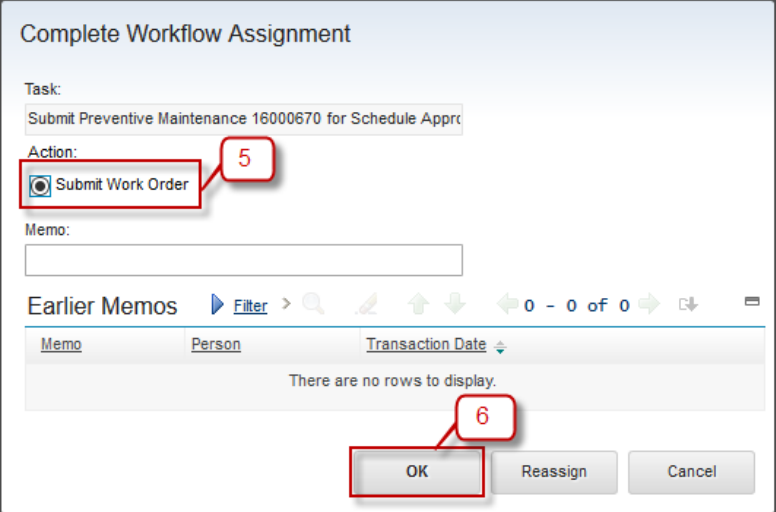
Is Rework?

☐

Originating Record:

Has Follow-up Work?

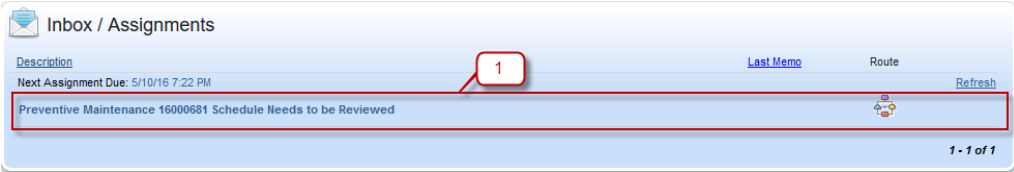
☐

5. Click Submit Work Order 6. Then click OK	
---	--

2.2.2 PM Work Schedule Approval by Operation

Below are steps to approve a PM work schedule by *Operation*. For rejection, revision request, and revision will not be discussed. They can refer to rejection, revision request, and revision of **SM Work Cycle** in step point **1.2.5 SM Request Revision Work Order** and **1.2.6 SM Revise and Re-submit Work Order** training material.

As a note, request revision for PM work schedule will direct the assignment back to *Service Engineer*. After revision is done, assignment will be directed back to *Operation* for schedule approval.

1. On Start Center of Operation, there will be on “Inbox / Assignment” then click the information 2. Click  for approval process	
3. Choose Approve Schedule 4. Click OK	

Complete Workflow Assignment

Task:

Preventive Maintenance 16000681 Schedule Needs to be Rev

Action:

☒ Approve Schedule
 ☐ Revise Schedule

Memo:

Earlier Memos

Filter > [Icons]

0 - 0 of 0

Memo	Person	Transaction Date
There are no rows to display.		

OK

Reassign

Cancel

2.2.3 PM Work Execution

PM work can be executed after the schedule is approved. To initiate the work, below are step by step needs to be done by *Service Engineer* on PM Work Order:

1. On Start Center Service Engineer there will be a new assignment in portlet **Inbox / Assignment**, choose the assignment

Inbox / Assignments

Next Assignment Due: 5/10/16 7:31 PM

1

Description	Last Memo	Start Date	Route	Refresh
Daily Inspection 16000661 Needs to be Executed		5/10/16 11:36 AM		
Submit for PM 1043		5/10/16 2:32 PM		
Daily Inspection 1034 Needs to be Closed		5/10/16 3:28 PM		
Submit 16000666 for Approval		5/10/16 4:37 PM		
Service Recovery 16000668 Needs to be Completed		5/10/16 6:13 PM		
Preventive Maintenance 16000670 Needs to be Completed		5/10/16 6:37 PM		
Submit Preventive Maintenance 16000685 for Schedule Approval		5/10/16 7:20 PM		
Preventive Maintenance 16000681 Needs to be Executed		5/10/16 7:31 PM		

<< Previous Page | 11 - 18 of 18

2. Choose **Execute Work Order**
3. Click **OK**

Complete Workflow Assignment

Task:
Preventive Maintenance 16000681 Needs to be Executed

Action:
☒ Execute Work Order

Memo:

Earlier Memos [Filter](#) >

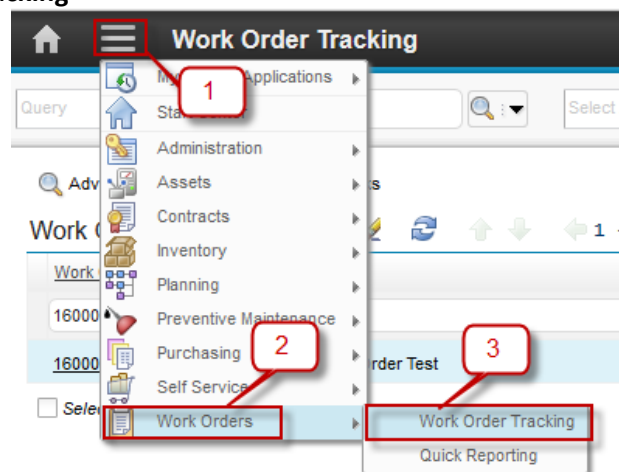
0 - 0 of 0

Memo	Person	Transaction Date
There are no rows to display.		

2.2.4 PM Work Completion

When PM work is finished, *Service Engineer* should input the work result in the PM Work Order. After that, the work order must be submitted for closing. Below are step by step to complete PM Work Order:

1. Click **Go To**
2. Navigate to **Work Order**
3. Click **Work Order Tracking**



4. Search **Work Order** on list **Work Order**
5. Then choose the **Work Order** no.

Work Order Tracking

Service Engineer01

Query: Find Work Order Select Action

Advanced Search Bookmarks

Work Orders Filter 1 - 20 of 171

Work Order	Description	Location	Asset	Status	Scheduled Start	Priority	Work Type	Site
1002				APPR				NPCT1
1003				APPR				NPCT1
1008	MM			APPR				NPCT1
1011	work order	NP1WS	NP1QCC101	WAPPR		CM		NPCT1
1018	Contoh CM 2	NP1BERTH	NP1QCC101	COMP	1/1/16 6:10 PM	CM		NPCT1
1022	Contoh AR - 1	NP1BERTH	NP1QCC101	WAPPR		AR		NPCT1
1028	Sample AR 01	NP1BERTH	NP1RTG202	APPR		AR		NPCT1
1032	Service Recovery Sample 01	NPCT1-ENG-STORE		INPRG				NPCT1
1034	Sample PM RTG 05	NP1BERTH	NP1RTG201	INPRG		DI		NPCT1
1042	Sample PM	NP1BERTH	1001	WEXEC		DI		NPCT1
1050	Sample AR 01	NP1BERTH	1001	WAPPR		AR		NPCT1
1053	Sample AR 03	NP1BERTH	1001	WAPPR		AR		NPCT1
1054	Sample SR 01	NP1		INPRG				NPCT1
1057	Sample Quick Reporting SR 2	NP1		COMP		SR		NPCT1
1060	Sample Quick Reporting SR 0 Sekian	NPCT1-ENG-STORE		INPRG		SR		NPCT1

6. Then click  for **Route Workflow**

7. Click  for **Complete WO**

8. Choose **Complete Work Order**

9. Click **OK**

Complete Workflow Assignment

Task: Preventive Maintenance 1 Needs to be Completed

Action: ☒ Complete Work Order

Memo:

Earlier Memos Filter 0 - 0 of 0

Memo	Person	Transaction Date
There are no rows to display.		

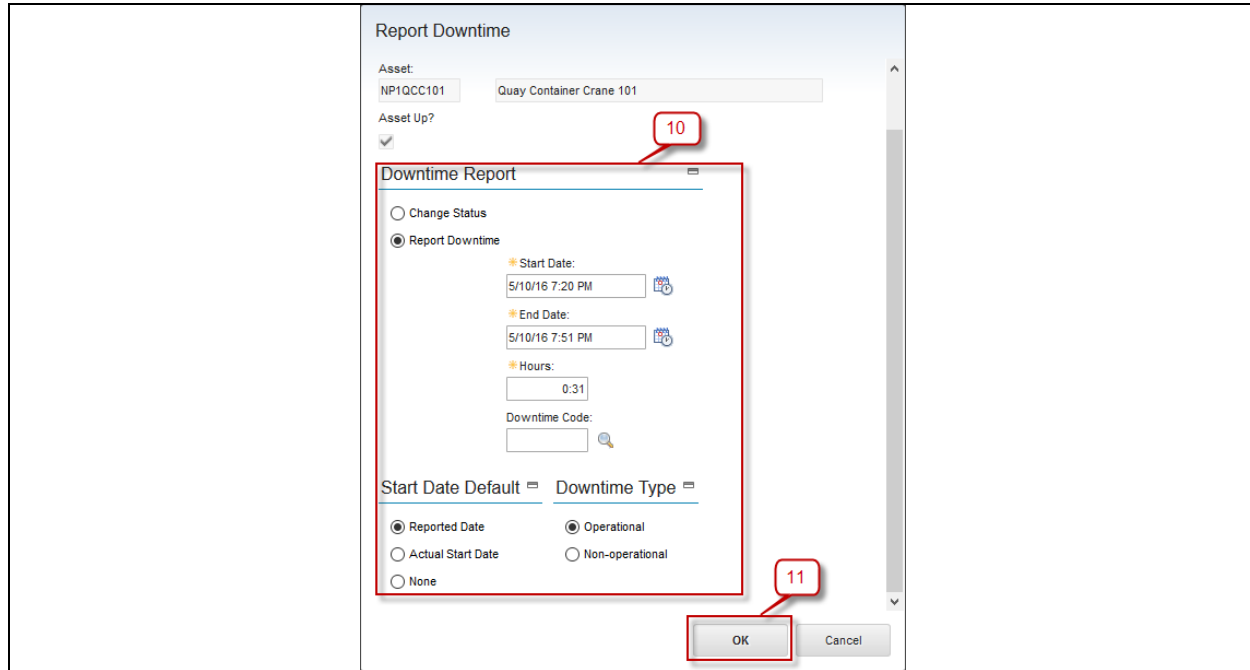
OK Reassign Cancel

10. Define a Downtime Report, such as: Start Date, End Date, Hours

11. Click **OK**

12. Then Status of **Work Order** has been changed to **COMP (Completed)**

13. Then click 



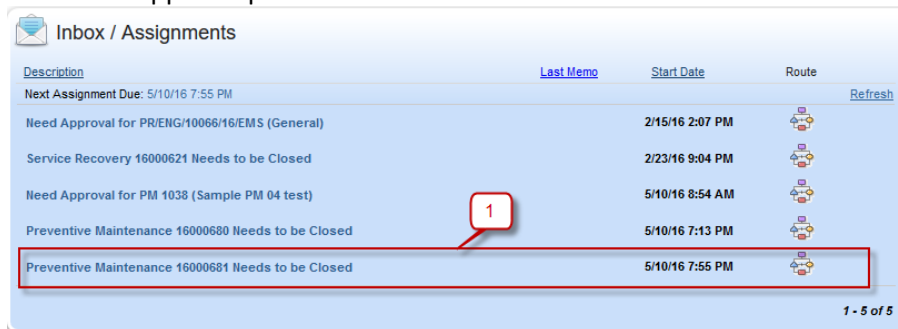
2.2.5 PM Work Closing Approval by CHE Engineer

Below are steps to approve a PM work closing by *CHE Engineer*. For rejection, revision request, and revision will not be discussed. They can refer to rejection, revision request, and revision of **SM Work Cycle** in step point **1.2.5 SM Request Revision Work Order** and **1.2.6 SM Revise and Re-submit Work Order** training material.

As a note, request revision for PM work closing will direct the assignment back to *Service Engineer*. After revision is done, assignment will be directed back to *CHE Engineer* for closing approval.

1. On CHE Start Center, in **Inbox / Assignment** portlet, click PM number to be Closed

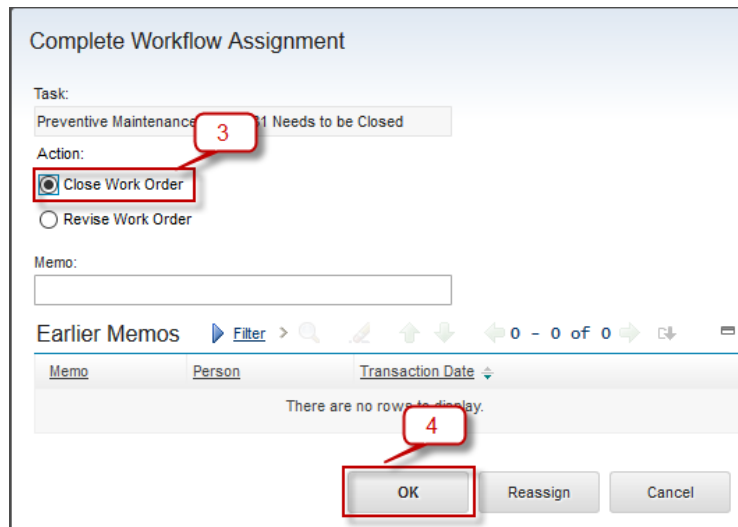
2. Then click  for approval process



Description	Last Memo	Start Date	Route
Next Assignment Due: 5/10/16 7:55 PM			
Need Approval for PR/ENG/10066/16/EMS (General)		2/15/16 2:07 PM	
Service Recovery 16000621 Needs to be Closed		2/23/16 9:04 PM	
Need Approval for PM 1038 (Sample PM 04 test)		5/10/16 8:54 AM	
Preventive Maintenance 16000680 Needs to be Closed		5/10/16 7:13 PM	
Preventive Maintenance 16000681 Needs to be Closed		5/10/16 7:55 PM	

3. Choose **Close Work Order**
4. Click **OK**

5. Then **Work Order** status has been **CLOSE**



Complete Workflow Assignment

Task:
Preventive Maintenance 1 Needs to be Closed

Action:
☒ Close Work Order
☐ Revise Work Order

Memo:

Earlier Memos [Filter](#) > [Search](#) [Refresh](#) [Export](#) 0 - 0 of 0

Memo	Person	Transaction Date
There are no rows to display.		

OK Reassign Cancel

2.3 Exercises

3 Daily Inspection (DI) Work

3.1 DI Work Overview

Daily Inspection (DI) work is a routine inspection to check assets or locations readiness for operation. DI work performed daily, before and after an assets or locations operates. When there is failure findings from this DI work, follow up work can be initiated. DI work order initiated based by PM schedule defined in **Preventive Maintenance** application.

A series of checklist should be presented in a DI work order, to guide the executor of what should be checked for specific asset or location. This can be accommodated by registering job plan containing those checklist in the PM schedule that will be applied automatically when the PM schedule creates DI work order.

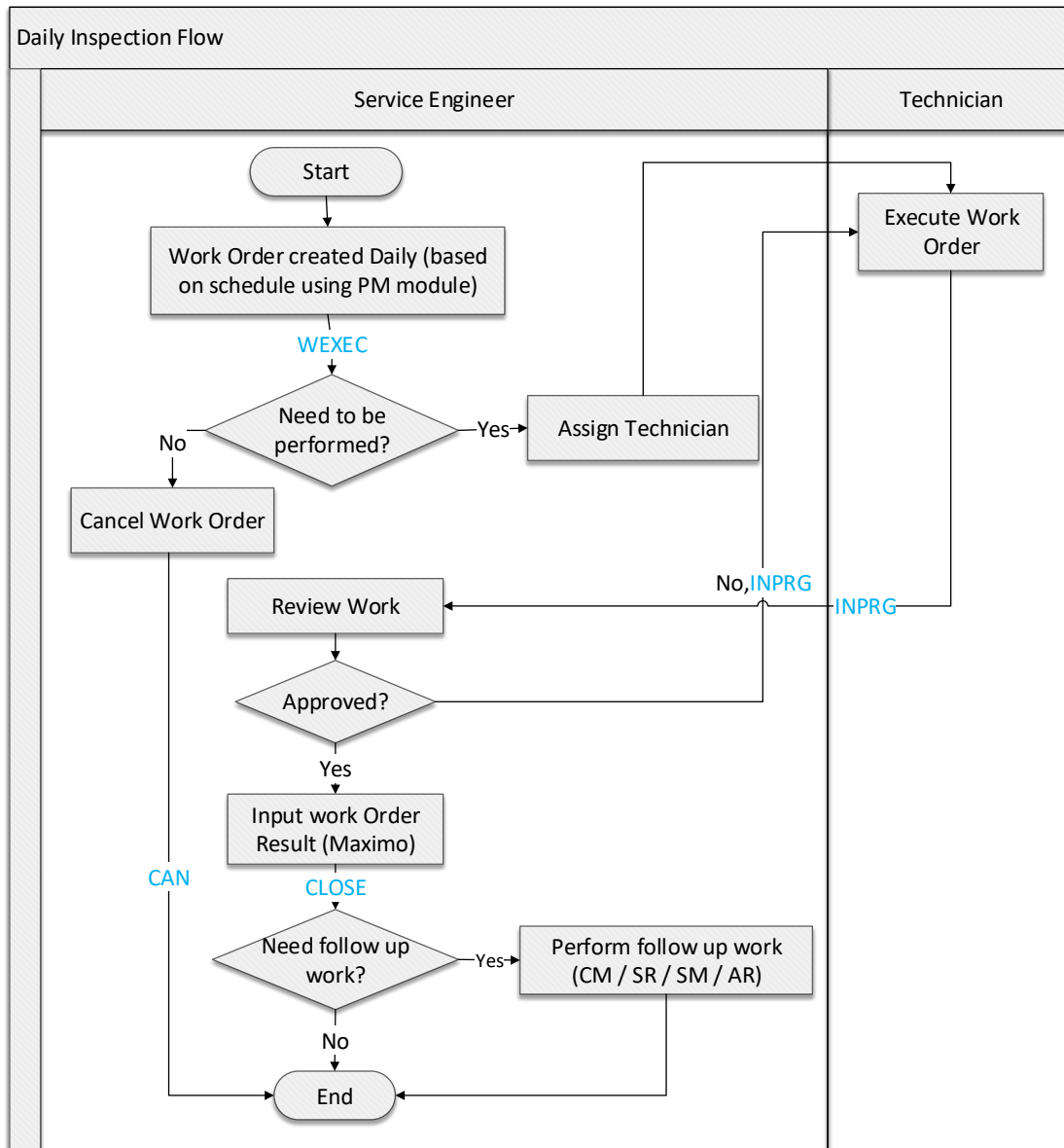
Sometimes DI work order cannot be performed on specific day. There are several cases that causes it, such as:

- The asset or location needs to operate immediately.
- Shortage of labor resources.
- Etc.

When those cases happen and prevent DI work to be executed, the DI work order can be cancelled.

3.2 DI Work Cycle

Below image show how Daily Inspection process flow managed in EMS.



Detail scenario for Daily Inspection (DI) work order:

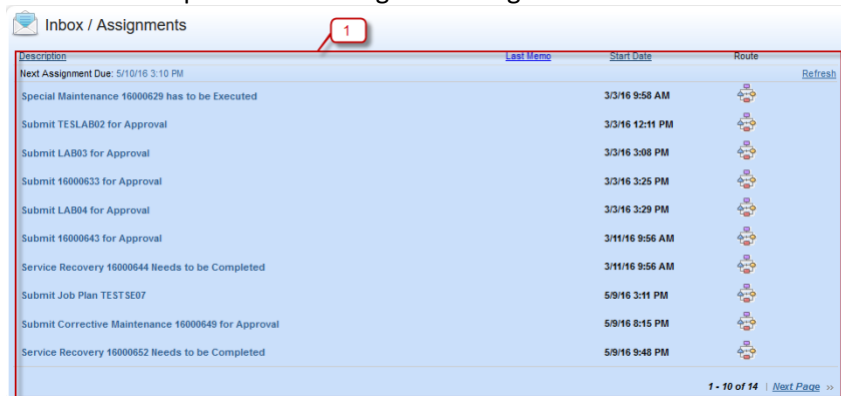
1. DI work order created automatically based on PM schedule defined in **Preventive Maintenance** application with the initial status is **WEEXEC**.
2. Service Engineer can cancel the work order if it does not need to be executed.
3. Service Engineer reviews the work result, input the result into Maximo, and close the work order after the work finish.
4. Failure findings gathered from this inspection can be followed up by creating a follow up work order (CM, SR, or SM).

3.2.1 DI Work Order Execution

DI Work Order generated based on PM schedule defined in **Preventive Maintenance** application. When a DI Work Order is generated, an assignment will appear in *Service Engineer's* Start Center. *Service Engineer* need to review the work order first, whether it can be executed or not.

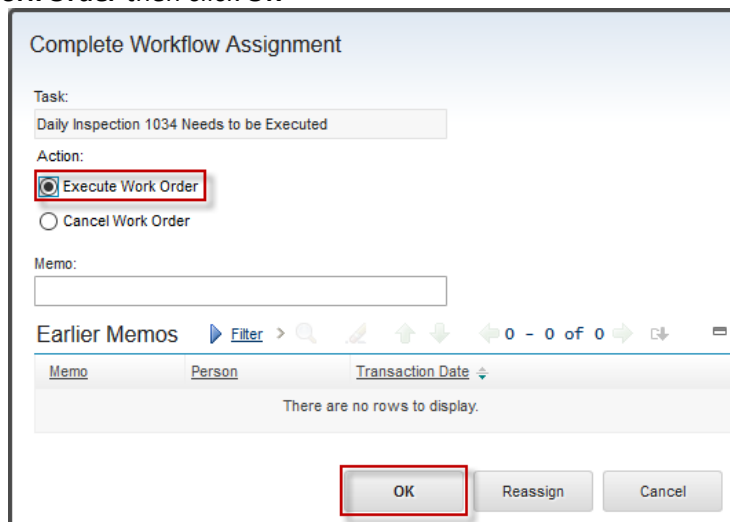
Below are step by step needs to be done by *Service Engineer* on DI work order to execute DI work:

1. Click Description of the Assignment to go to related Work Order



2. Click (Route Workflow)

3. Click **Execute Work Order** then click **OK**



Complete Workflow Assignment

Task:
Daily Inspection 1034 Needs to be Executed

Action:
☒ Execute Work Order
☐ Cancel Work Order

Memo:

Earlier Memos [Filter](#) >  0 - 0 of 0  

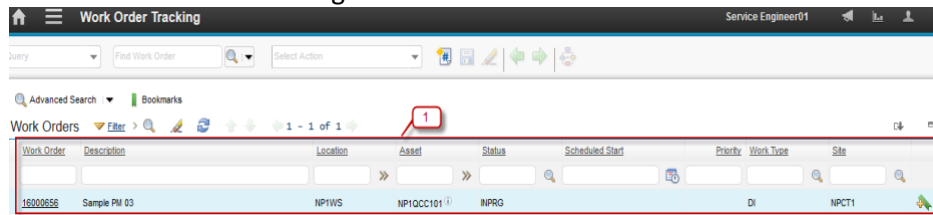
Memo	Person	Transaction Date
There are no rows to display.		

OK Reassign Cancel

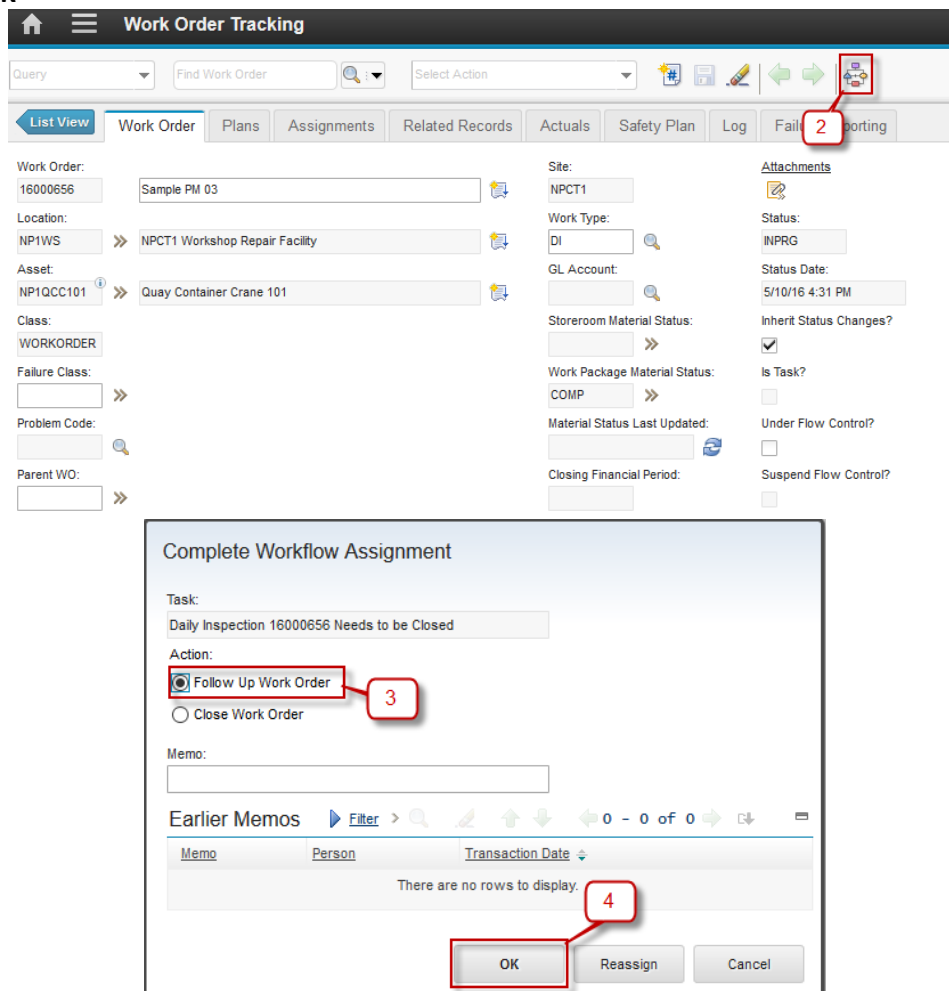
3.2.2 DI Work Completion

On DI work completion, *Service Engineer* should input the work result in the DI Work Order. After that, *Service Engineer* can decide whether follow up work order is needed or not.

1. Click on the list Work Order Tracking. Search and click the WO number



2. Click on  (Route Workflow)
3. Choose “Follow up Work Order”
4. Click OK



Work Order Tracking

Query: Find Work Order Select Action

Work Orders: 1 - 1 of 1

Work Order	Description	Location	Asset	Status	Scheduled Start	Priority	Work Type	Site
16000656	Sample PM 03	NP1WS	NP1QCC101	INPRG			DI	NPCT1

Work Order Details:

Work Order: 16000656 Sample PM 03

Location: NP1WS NPCT1 Workshop Repair Facility

Asset: NP1QCC101 Quay Container Crane 101

Class: WORKORDER

Failure Class:

Problem Code:

Parent WO:

Site: NPCT1

Work Type: DI

GL Account:

Storeroom Material Status:

Work Package Material Status: COMP

Material Status Last Updated:

Closing Financial Period:

Attachments:

Status: INPRG

Status Date: 5/10/16 4:31 PM

Inherit Status Changes? ☒

Is Task? ☐

Under Flow Control? ☐

Suspend Flow Control? ☐

Complete Workflow Assignment

Task: Daily Inspection 16000656 Needs to be Closed

Action:

☒ Follow Up Work Order

☐ Close Work Order

Memo:

Earlier Memos: 0 - 0 of 0

There are no rows to display.

OK Reassign Cancel